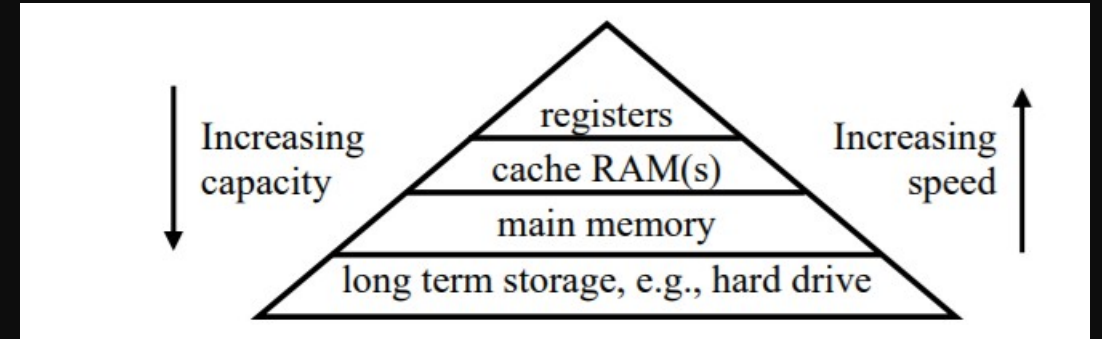


Lecture 8

Topics: Memory Hierarchy



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CIS-310: Computer Organization & Assembly Language

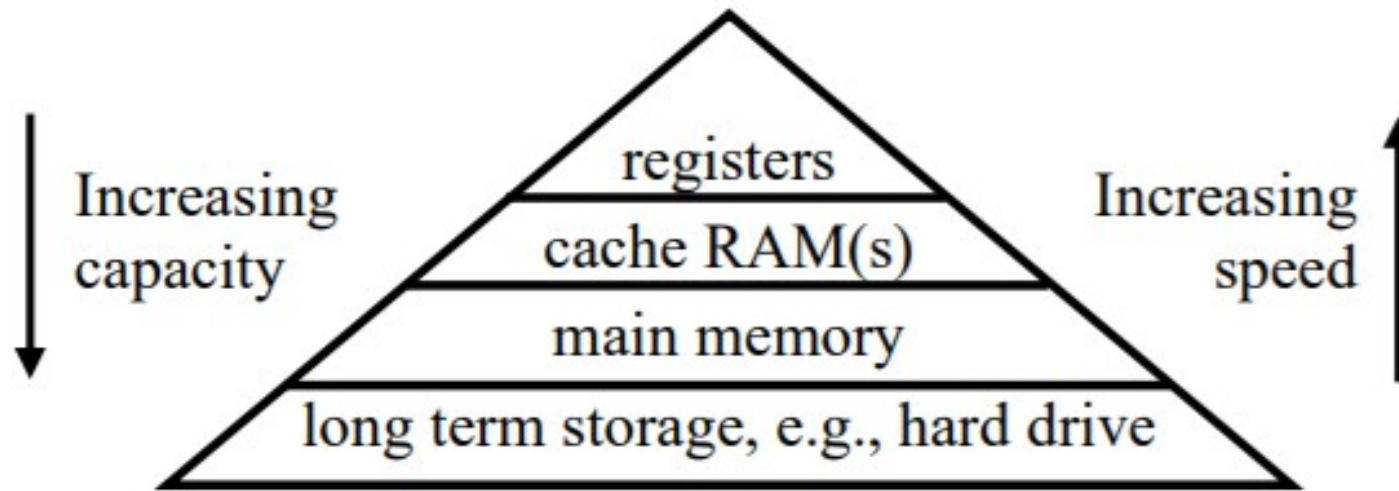
MW



So far:

- Memory construction
- Types of memory : SRAM, DRAM, ROM ...
- How CPU communicates with memory
- Memory mapping
- Memory addressing
- Address decoding

Block Diagram of a Standard Memory Hierarchy

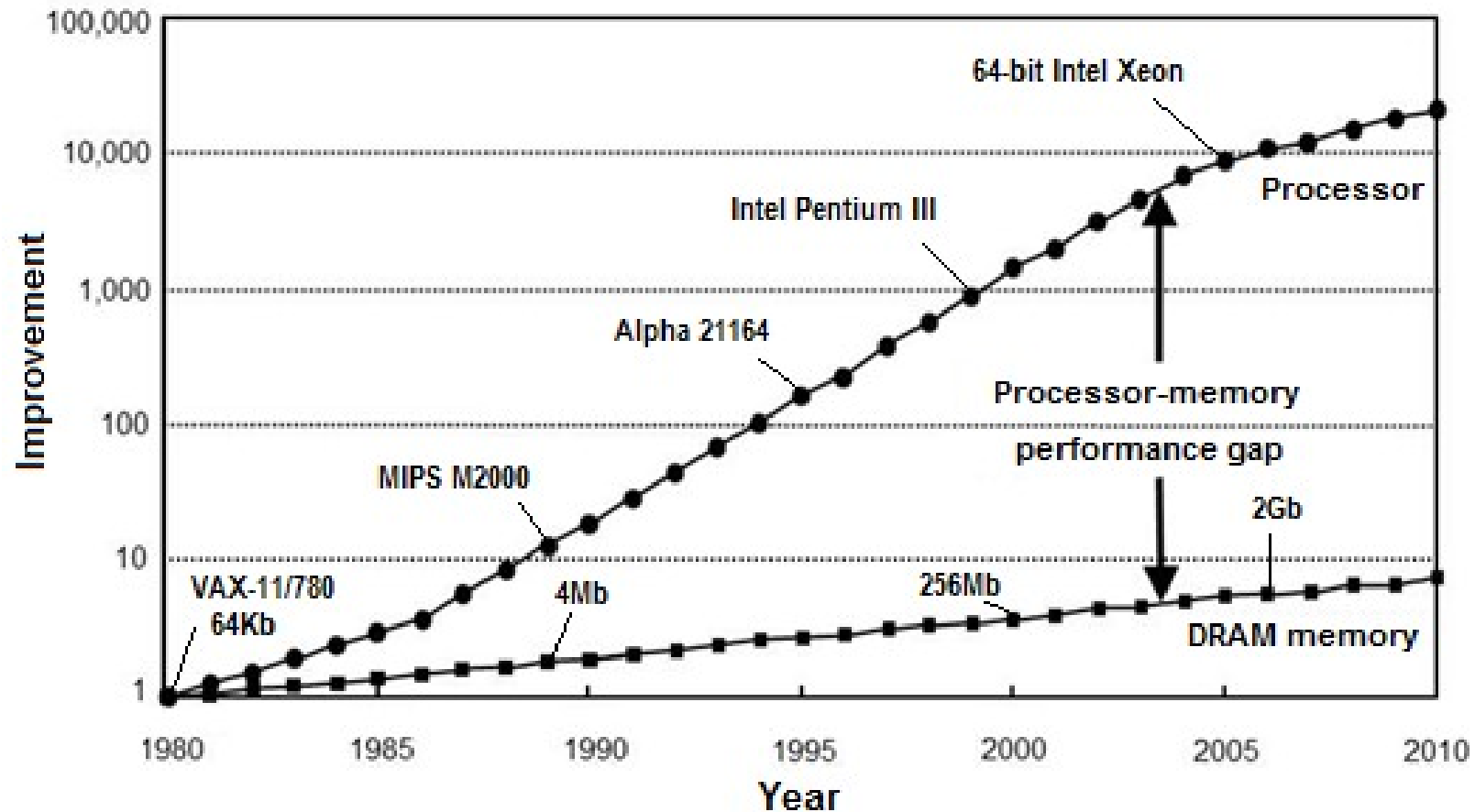


https://faculty.etsu.edu/tarnoff/ntes2150/Ch13_v02.pdf

Memory hierarchy

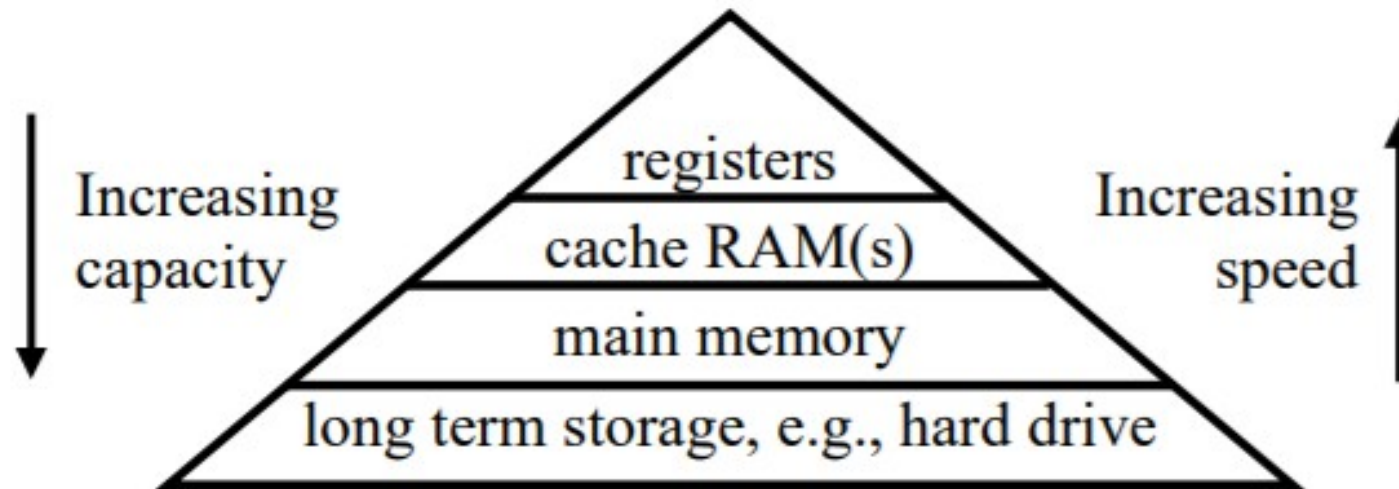
- A **memory hierarchy** is an organization of storage devices that takes advantage of the characteristics of different storage technologies in order to improve the overall performance of a computer system

Processor vs DRAM speed



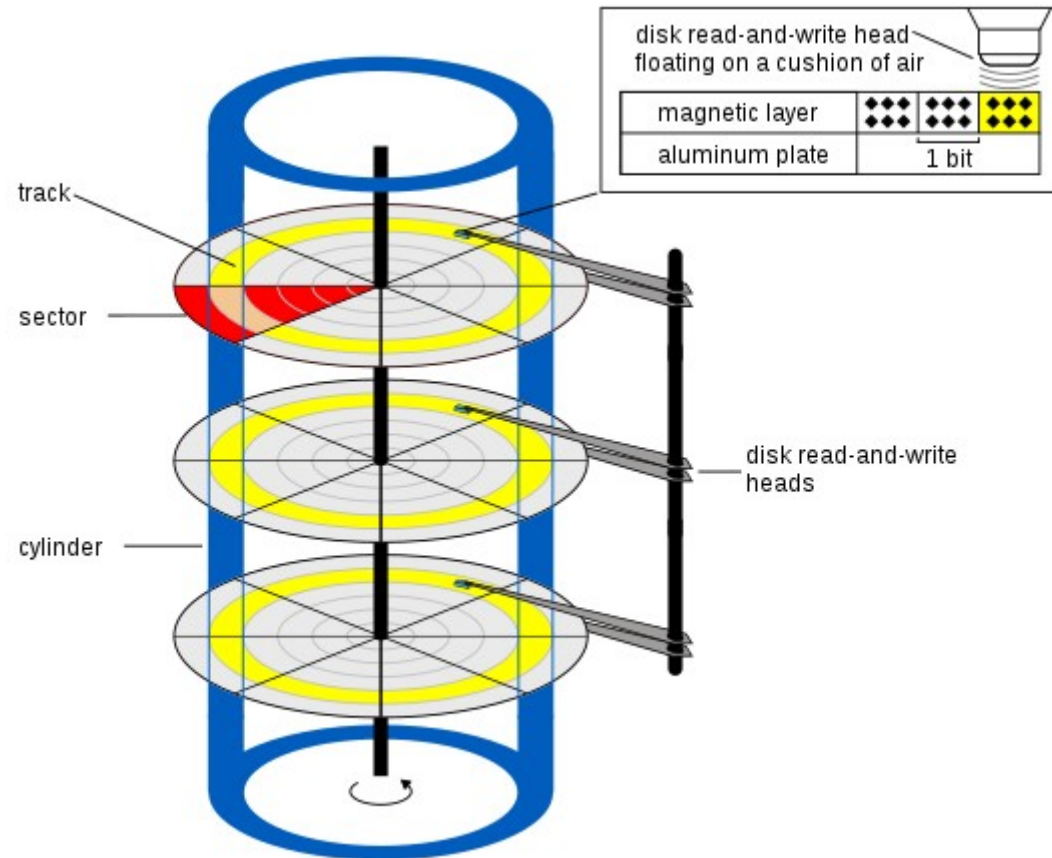
Goals of memory hierarchy

- Reduce cost
- Reduce access latency



Hard Drive

- <https://www.youtube.com/watch?v=NtPc0jI21i0>



Hard Drive Access Time

- There are four basic aspects to hard drive access time
 - queuing time,
 - seek time,
 - rotational latency,
 - and transfer time

Queuing time

- After an initial request is made to a hard drive, the system must wait for the hard drive to become available. This is called queuing time.
 - The hard drive may be busy serving another request or the bus
 - I/O channel that the hard drive uses may be busy serving another device that shares the link
 - The system's energy saving features may have powered down the drive

Seek time

- Seek time is the amount of time it takes to get the read/write head from its current track to the desired track
- It depends on:
 - The distance between the current and desired tracks
 - Mechanical movement of any sort requires a ramping up before attaining maximum speed and a ramping down to avoid overshooting the desired target position.

Rotational latency

- Once the head has been positioned over the desired track, the drive must wait for the platters to rotate to the sector containing the requested data. This is called rotational latency.
- The worst case occurs when the start of the desired sector has just passed under the head when the drive begins looking for the data. This requires almost a full rotation of the platters before the drive can begin transferring the data.
- We can use the following calculation to determine the time required for a platter in a 7200 RPM drive to make a full rotation

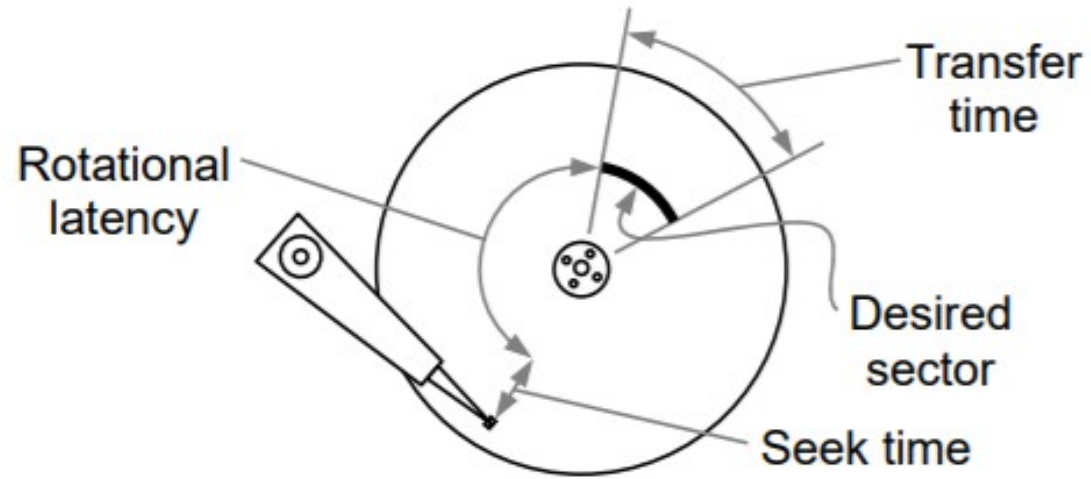
$$\frac{1 \text{ minute}}{7200 \text{ rotations}} \times \frac{60 \text{ seconds}}{1 \text{ minute}} = 8.3 \text{ ms per rotation}$$

Transfer time

- Transfer time is the time it takes to send the requested data from the hard drive to the requesting device.
- The maximum transfer time equals the amount of time it takes for the data to pass beneath the head.
- If there are N sectors per track, then the amount of time it takes to retrieve a single sector can be calculated as shown below

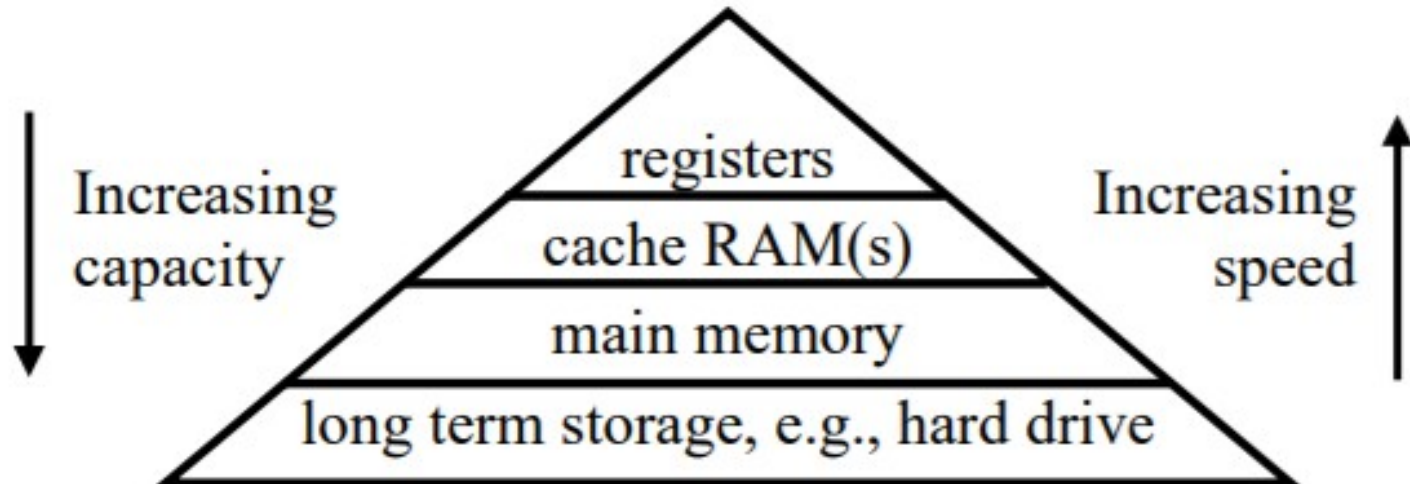
$$\text{Theoretical transfer time for a sector} = (N \times \text{rotational speed})^{-1}$$

Disk access time



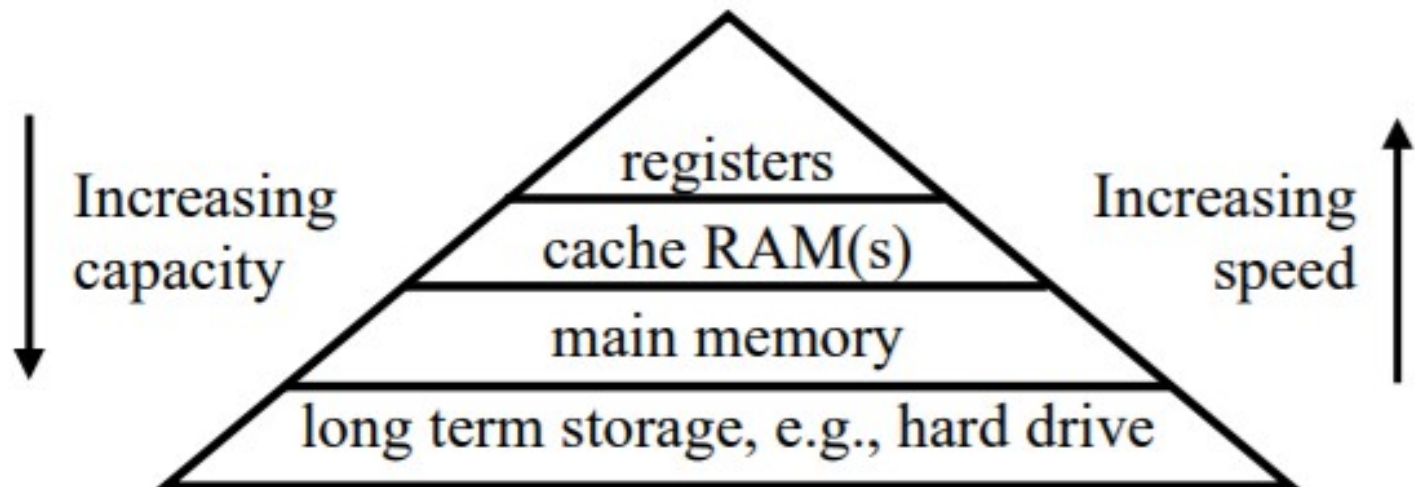
Cons of hard drive

- Even with increases in hard drive performance, it will never be practical to execute programs or access data directly from these mechanical devices.
 - . They are far too slow



Solution?

- When the processor needs to access information, it is first loaded from the hard drive into main memory where the higher performance RAM allows fast access to the data. When the processor is finished with the data, the information can either be discarded or used to update the hard drive.



RAM – Random access memory

- There are two main classifications of RAM:
 - DRAM: Dynamic RAM
 - SRAM: Static RAM

<u>SRAM</u>	<u>DRAM</u>
1. SRAM has lower access time, so it is faster compared to DRAM.	1. DRAM has higher access time, so it is slower than SRAM.
2. SRAM is costlier than DRAM.	2. DRAM costs less compared to SRAM.
3. SRAM requires constant power supply, which means this type of memory consumes more power.	3. DRAM offers reduced power consumption, due to the fact that the information is stored in the capacitor.
4. Due to complex internal circuitry, less storage capacity is available compared to the same physical size of DRAM memory chip.	4. Due to the small internal circuitry in the one-bit memory cell of DRAM, the large storage capacity is available.
5. SRAM has low packaging density.	5. DRAM has high packaging density.

Cache RAM



Figure 13-13 Cache Placement between Main Memory and Processor

Multi-level cache



Data locality

- <https://gist.github.com/proywm/83d1d4580b5589a207c5a8d917e797da>

Amazon study: Every 100ms in Added Page Load Time Cost 1% in Revenue

Back in 2006, Amazon found that every 100ms in added page load time cost them 1% in sales. This has now become one of the most referenced data points surrounding page speed and web performance — standing the test of time as a clear example for why having a fast site is important.

For context, a 1% loss of [annual revenue for Amazon](#) in 2006 would have been around \$107 million. Today, this would be about \$3.8 billion!

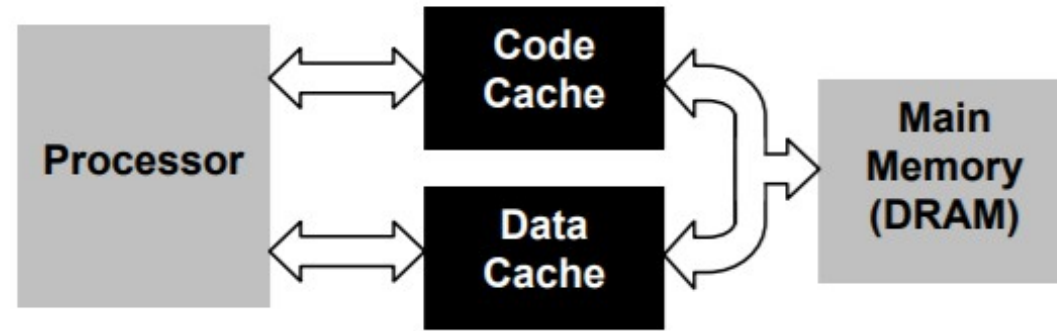


Figure 13-15 Split Cache Organization

Cache Organization

Dividing Memory into Blocks

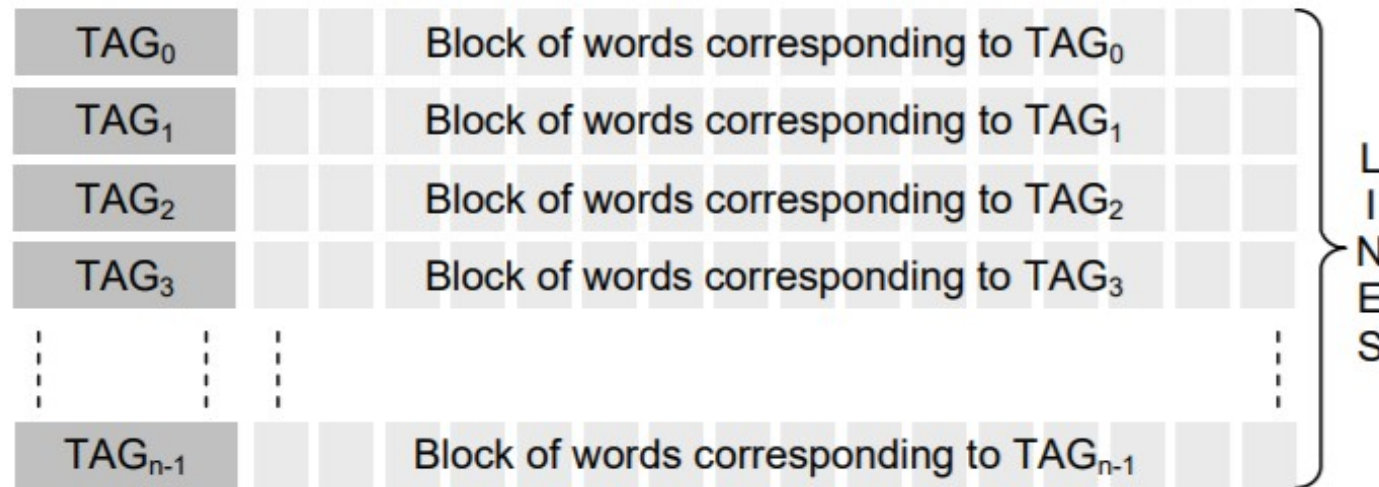
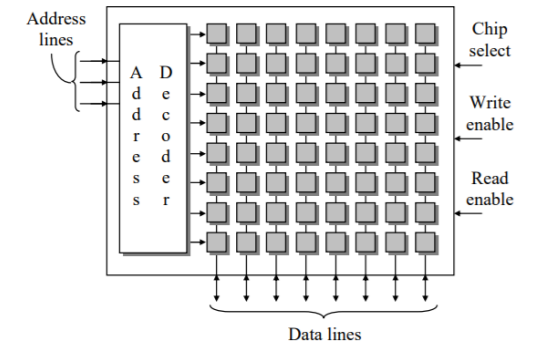


Figure 13-16 Organization of Cache into Lines

Division of Memory into Blocks

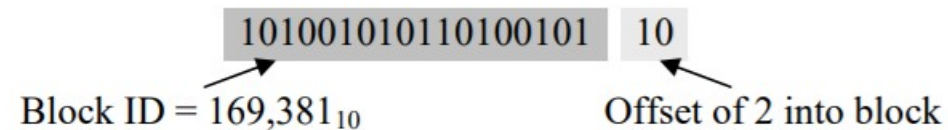
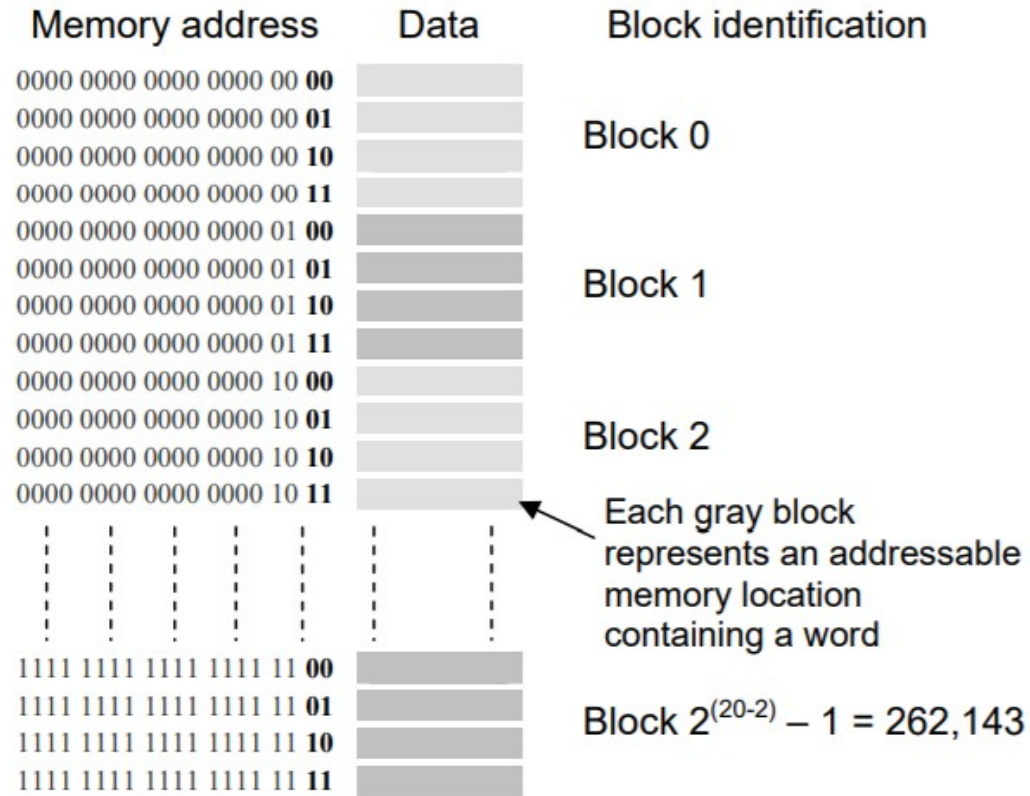


figure 13-18 Organization of Address Identifying Block and Offset

Cache Characteristics

- The cache system used by a processor is defined by six traits:
 - the size of the cache;
 - the size of a block, which when combined with the size of the cache defines the number of lines;
 - the number of caches (i.e., multiple levels or a split cache);
 - the mapping function (the link between a block's address in memory and its location in the cache);
 - the replacement algorithm (the method used to figure out which block to remove from the cache in order to free up a line); and
 - the write policy (how the processor writes data to the cache so that main memory eventually gets updated)